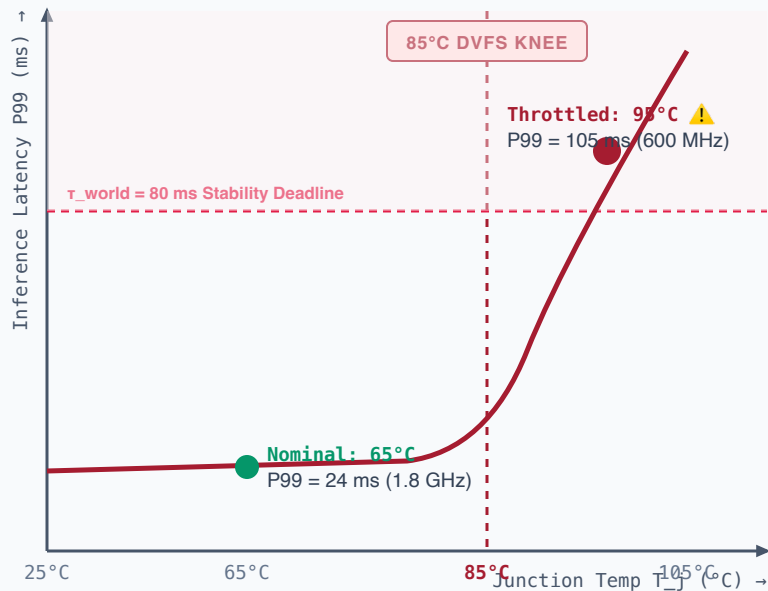


THERMAL DERATING CLIFF & THE RE-PLACEMENT RIPPLE EFFECT

Silicon DVFS throttling crossing the 85°C junction knee: latency tripling, deadline breach, and stopping distance expansion

SILICON JUNCTION TEMPERATURE VS. INFERENCE TAIL LATENCY



PHYSICAL RE-PLACEMENT RIPPLE EFFECT

KINETIC CONSEQUENCE: STOPPING DISTANCE COLLAPSE

~~$d_{stop} = v_{max} (t_{latency} + t_{brake}) + v^2 / (2 a_{max})$~~

$d_{stop} = 0.89 \text{ m}$ COLLISION

• Thermal throttle causes **134% increase in stopping distance**, breaching physical safety clearance

ADAPTIVE RE-PLACEMENT CODESIGN STRATEGIES

- Dynamic Model Switching:** Shift from heavy ViT-Large to quantized INT4 MobileNet to stay within thermal envelope
- Adaptive Velocity Scaling:** Reduce robot max cruise speed v_{max} proportionally to measured latency increase
- Cloud Offloading with Graceful Fallback:** Offload System 2 deliberation to off-board servers via 5G edge network